# Basics of SAP and ERP

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## ERP

Enterprise Resource Planning
Purpose is to manage any company functions.

## SAP

System Application and Products for Real Time Data processing.

## Features of SAP

- Integration of all functions in one common software.
- Multi-lingual software.
- User-based license agreement.

## SAP R/3 Architecture

(3 tier Architecture, R- Real ) - SAP is real time architecture.
Layers - Presentation Layer (user Interaction Layer).
Application Layer (writing programs and applications)
Database Layer (to store and retrieve data)

## ABAP / 4

Advanced Business Application Programming language
ABAP lies in application layer.

## Modules of SAP

Functional Module
Technical Modules

## Functional Modules

- SAP SD (Sales & Distribution)
- SAP MM (Material Management)
- SAP PS (Project System)
- SAP FICO (Finance & Controlling)
- SAP HR (Human Capital Management)
- SAP WM (Warehouse Management)

## Technical Modules

- SAP BASIS -> Administration and Monitoring of SAP System.
- SAP ABAP -> Programming language for SAP.
- SAP BIBO -> Reporting Tool

## SAP NetWeaver

SAP NetWeaver is an integrated technology platform which support various SAP applications.
SAP ERP (Enterprise Resource Planning)
SAP CRM (Customer Relationship Management)
SAP SRM (Supply Relationship Management)
SAP PI (Process Integration)

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## Application Server Architecture

1. Dispatcher
2. Work Process
   a. User Context
      i. Memory area that contains info about user.
      ii. User logon -> user context is allocated
      iii. User log off -> memory freed
   b. Roll Area
      i. Memory area that contains information about program execution
      ii. Program runs -> roll area allocated
      iii. Program ends -> roll area freed.

## SAP System landscape

- Development -> Dedicated for development
- Quality -> Testing
- Production -> Used by customer

*Transport system : Mechanism to transport object from one system to another system*

## Types of SAP GUI

- SAP GUI for Windows - Windows OS
- SAP GUI for Java - Written in Java and is platform-independent implementation of SAP-GUI.
- SAP GUI for HTML - Consists of Internet Transaction Server on server side and web browser on client side.

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## ABAP Workbench

Collection of ABAP tools
SE38 -> Program/Report
SE11 -> ABAP Dictionary
SE37 -> Function Module
SE24 -> Class

## Transaction Code

Shortcut to access ABAP Workbench tools
/Nxxx - to call Transaction xxx (overide)
/Oxxx - to open separate Transaction xxx
**User can open maximum of six session**

F4 Help -> Tells us what are various possible values for any input field.
F1 Help -> Gives us technical information of any field/column.

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## SAP Specific Objects & Customer Specific Objects.

Customer Specific Objects starts with Z or Y.
SAP Specific Objects starts with other than Z or Y.

## Object Navigator SE80

SE80 --> Object navigator

## Package SE21

Collection for the objects (Table, Data element, Domains, Structures , Programs etc...)
In SAP every object stored into package.

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## Header and Item Tables

VBAK - Sales Order Header Data
VBAP - Sales Order Item Data

MARA - Material Data
MAKT - Material Description

## Transaction Codes - Sales & Distribution SD

- VA01 - Sales order Create
- VA02 - Sales order Change
- VA03 - Sales order Display

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## Types of Projects in SAP

- Implementation - Customer is implementing SAP for first time. All legacy system data needs to be migrated to SAP.
- Support - Also called Maintenance project. Customer is running his business through SAP. Customer requires support for daily issues an d new change requests.
- Upgrade - Customer moving from older version of SAP to higher version of SAP.
- Roll Out - Existing SAP system is going to implement for different countries/regions.

## RICEFW

### Reports

- Used to display of information in required format.
- If business requirement cannot be fulfilled by standard SAP reports then develop custom reports based upon customer requirements.
- Various report formats are classical reports, ALV reports.

### Interface

- Interface are used to transfer data from Non-SAP to SAP, SAP to Non-SAP
- Various cross-applications like BAPI, RFC, IDOC's are examples of interfaces.

## Conversion

- Process of converting data from one form to another is called conversion.
- When customer is implementing SAP for first time, all legacy system data needs to be migrated to SAP system.
- To upload data in SAP system data needs to be converted from one form to another as per system requirement.
- After conversion data is uploaded into SAP system by making use of various data migratIon tools like BDC, LSMW, BAPI

## Enhancement

- Process of adding own functionality to SAP standard functionality is called enhancements.
- If business requirement cannot be satisfied by standard sap functionality, then we add our own functionality to SAP standard functionality.

## Forms

- Forms are layouts in SAP.
- If business requirement cannot be satisfied by standard layouts, then we create our own custom layout.
- We design layout in SAP by using various tools like SMARTFORMS, adobe forms, SAP scripts.

## Workflow.

- Workflow is flowing of data from one level to another level.
- Eg: when we apply leave in system, it goes into workflow of supervisor to approve or reject leave.

## RICEFW / WRICEF

SAP technical object into 6 categories

| | |
|---|---|
| Reports | R |
| Interface | I |
| Conversion | C |
| Enhancement | E |
| Forms | F |
| Workflow | W |

## ECC

ERP Central Component
Application Layer - ECC
Database layer - oracle, Sybase, DB2

## SOH

SAP Business Suite on HANA
Uses only database HANA
Application Layer - ECC
Database Layer - HANA

## S4H / S4 HANA

Complete suite is in HANA
Application layer - HANA
Database layer - HANA